

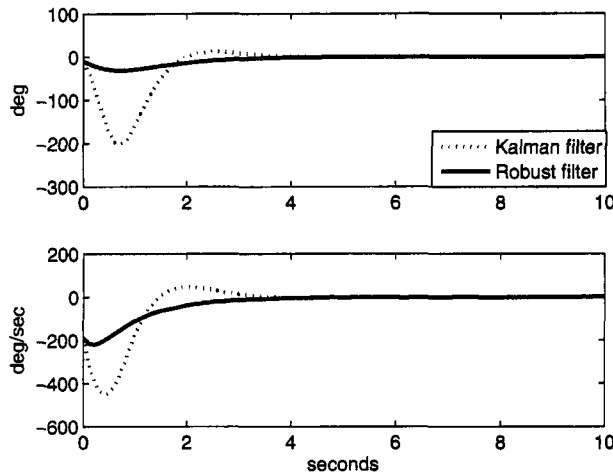


Figure 12.3 Position and velocity estimation errors for Example 12.2 for the Kalman filter and the robust filter, assuming that the system model is not well known. The robust filter performs better both at the beginning of the simulation and in steady state. The steady-state Kalman filter estimation errors have standard deviations of 0.36 deg and 1.33 deg/s, whereas the robust filter has standard deviations of 0.28 deg and 1.29 deg/s.

12.3 CONSTRAINED H_∞ FILTERING

As in Section 7.5, suppose that we know (on the basis of physical considerations) that the states satisfy some equality constraint $D_k x_k = d_k$, or some inequality constraint $D_k x_k \leq d_k$, where D_k is a known matrix and d_k is a known vector. This section discusses how those constraints can be incorporated into the H_∞ filter equations. As discussed in Section 7.5, state equality constraints can always be handled by reducing the system-model parameterization [Wen92], or by treating state equality constraints as perfect measurements [Por88, Hay98]. However, these approaches cannot be extended to inequality constraints. The approach summarized in this section is to incorporate the state constraints into the derivation of the H_∞ filter [Sim06c].

Consider the discrete LTI system given by

$$\begin{aligned} x_{k+1} &= Fx_k + w_k + \delta_k \\ y_k &= Hx_k + v_k \end{aligned} \quad (12.31)$$

where y_k is the measurement, $\{w_k\}$ and $\{v_k\}$ are uncorrelated white noise sequences with respective covariances Q and I , and $\{\delta_k\}$ is a noise sequence generated by an adversary (i.e., nature). Note that we are assuming that the measurement noise has a unity covariance matrix. In a real system, if the measurement noise covariance is not equal to the identity matrix, then we will have to normalize the measurement equation as shown in Example 12.3 below. In general, F , H , and Q can be time-varying matrices, but we will omit the time subscript on these matrices for ease of notation. In addition to the state equation, we know (on the basis of physical considerations or other *a priori* information) that the states satisfy the following

constraint:

$$D_k x_k = d_k \quad (12.32)$$

We assume that the D_k matrix is full rank and normalized so that $D_k D_k^T = I$. In general, D_k is an $s \times n$ matrix, where s is the number of constraints, n is the number of states, and $s < n$. If $s = n$ then Equation (12.32) completely defines x_k , which makes the estimation problem trivial (i.e., $\hat{x}_k = D_k^{-1} d_k$). For $s < n$, which is the case in this section, there are fewer constraints than states, which makes the estimation problem nontrivial. Assuming that D_k is full rank is the same as the assumption made in the constrained Kalman filtering problem of Section 7.5. For notational convenience we define the matrix V_k as

$$V_k = D_k^T D_k \quad (12.33)$$

We assume that both the noisy system and the noise-free system satisfy the state constraint. The problem is to find an estimate $\hat{x}_{k+1}$ of x_{k+1} given the measurements $\{y_1, y_2, \dots, y_k\}$. The estimate should satisfy the state constraint. We will assume that the estimate is given by the following standard predictor/corrector form:

$$\begin{aligned} \hat{x}_0 &= 0 \\ \hat{x}_{k+1} &= F \hat{x}_k + K_k (y_k - H \hat{x}_k) \end{aligned} \quad (12.34)$$

The noise δ_k in (12.31) is introduced by an adversary that has the goal of maximizing the estimation error. We will assume that our adversary's input to the system is given as follows:

$$\delta_k = L_k [G_k (x_k - \hat{x}_k) + n_k] \quad (12.35)$$

where L_k is a gain to be determined, G_k is a given matrix, and $\{n_k\}$ is a noise sequence with variance equal to the identity matrix. We assume that $\{n_k\}$ is uncorrelated with $\{w_k\}$, $\{v_k\}$, and x_0 . This form of the adversary's input is not intuitive because it is based on the state estimation error, but this form is taken because the solution of the resulting problem results in a state estimator that bounds the infinity-norm of the transfer function from the random noise terms to the state estimation error [Yae92].

G_k in Equation (12.35) is chosen by the designer as a tuning parameter or weighting matrix that can be adjusted on the basis of our *a priori* knowledge about the adversary's noise input. Suppose, for example, that we know ahead of time that the first component of the adversary's noise input to the system is twice the magnitude of the second component, the third component is zero, and so on; then that information can be reflected in the designer's choice of G_k . We do not need to make any assumptions about the form of G_k (e.g., it does not need to be positive definite or square). From Equation (12.35) we see that as G_k approaches the zero matrix, the adversary's input becomes a purely random process without any deterministic component. This causes the resulting filter to approach the Kalman filter; that is, we obtain better RMS error performance but poorer worst-case error performance. As G_k becomes large, the filter places more emphasis on minimizing the estimation error due to the deterministic component of the adversary's input. That is, the filter assumes less about the adversary's input, and we obtain better worst-case error performance but worse RMS error performance. The estimation error is defined as

$$e_k = x_k - \hat{x}_k \quad (12.36)$$

It can be shown from the preceding equations that the dynamic system describing the estimation error is given as

$$\begin{aligned} e_0 &= x_0 \\ e_{k+1} &= (F - K_k H + L_k G_k) e_k + w_k + L_k n_k - K_k v_k \end{aligned} \quad (12.37)$$

Since $D_k x_k = D_k \hat{x}_k = d_k$, we see that $D_k e_k = 0$. But it can also be shown [Sim06c] that $D_{k+1} F e_k = 0$. Therefore, we can subtract the zero term $D_{k+1}^T D_{k+1} F e_k = V_{k+1} F e_k$ from Equation (12.37) to obtain

$$\begin{aligned} e_0 &= x_0 \\ e_{k+1} &= [(I - V_{k+1})F - K_k H + L_k G_k] e_k + w_k + L_k n_k - K_k v_k \end{aligned} \quad (12.38)$$

However, this is an inappropriate term for a minimax problem because the adversary can arbitrarily increase e_k by arbitrarily increasing L_k . To prevent this, we decompose e_k as

$$e_k = e_{1,k} + e_{2,k} \quad (12.39)$$

where $e_{1,k}$ and $e_{2,k}$ evolve as follows:

$$\begin{aligned} e_{1,0} &= x_0 \\ e_{1,k} &= [(I - V_{k+1})F - K_k H + L_k G_k] e_{1,k} + w_k - K_k v_k \\ e_{2,0} &= 0 \\ e_{2,k} &= [(I - V_{k+1})F - K_k H + L_k G_k] e_{2,k} + L_k n_k \end{aligned} \quad (12.40)$$

We define the objective function for the filtering problem as

$$J(K, L) = \text{trace} \sum_{k=0}^N W_k E (e_{1,k} e_{1,k}^T - e_{2,k} e_{2,k}^T) \quad (12.41)$$

where W_k is any positive definite weighting matrix. The differential game is for the filter designer to find a gain sequence $\{K_k\}$ that minimizes J , and for the adversary to find a gain sequence $\{L_k\}$ that maximizes J . As such, J is considered a function of $\{K_k\}$ and $\{L_k\}$, which we denote in shorthand notation as K and L . This objective function is not intuitive, but is used here because the solution of the problem results in a state estimator that bounds the infinity-norm of the transfer function from the random noise terms to the state estimation error [Yae92]. That is, suppose we can find an estimator gain K^* that minimizes $J(K, L)$ when the matrix G_k in (12.35) is equal to θI for some positive scalar θ . Then the infinity-norm of the weighted transfer function from the noise terms w_k and v_k to the estimation error e_k is bounded by $1/\theta$. That is,

$$\sup_{w_k, v_k} \frac{\sum_{k=0}^N e_k^T e_k}{\sum_{k=0}^N (w_k^T Q^{-1} w_k + v_k^T v_k)} < \frac{1}{\theta} \quad (12.42)$$

where *sup* stands for *supremum*.¹ The filtering solution is obtained by finding optimal gain sequences $\{K_k^*\}$ and $\{L_k^*\}$ that satisfy the following saddle point:

$$J(K^*, L) \leq J(K^*, L^*) \leq J(K, L^*) \text{ for all } K, L \quad (12.43)$$

¹The supremum of a function is its least upper bound. This is similar to the maximum of a function, but a maximum is a value that is actually attained by a function, whereas a supremum may or may not be attained. For example, the supremum of $(1 - e^{-x})$ is 1, but $(1 - e^{-x})$ never actually reaches the value 1. Similar distinctions hold for the operators minimum and infimum (usually abbreviated *inf*).

This problem is solved subject to the constraint that $D_k \hat{x}_k = d_k$ in [Sim06c], whose result is presented here. We define P_k and Σ_k as the nonsingular solutions to the following set of equations:

$$\begin{aligned} P_0 &= E(x_0 x_0^T) \\ \Sigma_k &= (P_k H^T H - P_k G_k^T G_k + I)^{-1} P_k \\ P_{k+1} &= (I - V_{k+1}) F \Sigma_k F^T (I - V_{k+1}) + Q \end{aligned} \quad (12.44)$$

Nonsingular solutions to these equations are not always guaranteed to exist, in which case a solution to the H_∞ filtering problem may not exist. However, if nonsingular solutions do exist, then the following gain matrices for our estimator and adversary satisfy the constrained H_∞ filtering problem:

$$\begin{aligned} K_k^* &= (I - V_{k+1}) F \Sigma_k H^T \\ L_k^* &= (I - V_{k+1}) F \Sigma_k G_k^T \end{aligned} \quad (12.45)$$

These matrices solve the constrained H_∞ filtering problem only if $(I - G_k P_k G_k^T) \geq 0$. Note that as G_k becomes larger, we will be less likely to satisfy this condition. From Equation (12.35) we see that a larger G_k gives the adversary more latitude in choosing a disturbance. This makes it less likely that the designer can minimize the cost function.

The mean square estimation error that results from using the optimal gain K_k^* cannot be specified because it depends on the adversary's input δ_k . However, we can state an upper bound for the mean square estimation error [Sim06c] as follows:

$$E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] \leq P_k \quad (12.46)$$

This provides additional motivation for using the game theory approach presented in this section. The estimator not only bounds the worst-case estimation error, but also bounds the mean square estimation error.

Now consider the special case that there are no state constraints. Then in Equation (12.32) we can set the D_k matrix equal to a zero row vector and the d_k vector equal to the zero scalar. In this case $V_{k+1} = 0$ and we obtain from Equations (12.44) and (12.45) the following estimator and adversary strategies:

$$\begin{aligned} P_0 &= E(x_0 x_0^T) \\ P_k(I - H^T H \Sigma_k) &= (I - P_k G_k^T G_k) \Sigma_k \\ P_{k+1} &= F \Sigma_k F^T + Q \\ K_k^* &= F \Sigma_k C^T \\ L_k^* &= F \Sigma_k G_k^T \end{aligned} \quad (12.47)$$

This is identical to the unconstrained H_∞ estimator [Yae92]. The unconstrained H_∞ estimator for continuous-time systems is given in [Yae04].

In the case of state inequality constraints (i.e., constraints of the form $D_k x_k \leq d_k$), a standard active-set method [Fle81, Gil81] can be used to solve the H_∞ filtering problem. An active-set method uses the fact that it is only those constraints that are active at the solution of the problem that affect the optimality conditions; the inactive constraints can be ignored. Therefore, an inequality-constrained problem is equivalent to an equality-constrained problem. An active-set method determines

which constraints are active at the solution of the problem and then solves the problem using the active constraints as equality constraints. Inequality constraints will significantly increase the computational effort required for a problem solution because the active constraints need to be determined, but conceptually this poses no difficulty.

The constrained H_∞ filter

The constrained H_∞ filter can be summarized as follows.

1. We have a linear system given as

$$\begin{aligned} x_{k+1} &= F_k x_k + w_k \\ y_k &= H_k x_k + v_k \\ D_k x_k &= d_k \end{aligned} \quad (12.48)$$

where w_k is the process noise, v_k is the measurement noise, and the last equation above specifies equality constraints on the state. We assume that the constraints are normalized so $D_k D_k^T = I$. The covariance of w_k is equal to Q_k , but w_k might have a zero mean or it might have a nonzero mean (i.e., it might contain a deterministic component). The covariance of v_k is the identity matrix.

2. Initialize the filter as follows:

$$\begin{aligned} \hat{x}_0 &= 0 \\ P_0 &= E(x_0 x_0^T) \end{aligned} \quad (12.49)$$

3. At each time step $k = 0, 1, \dots$, do the following.

- (a) Choose the tuning parameter matrix G_k to weight the deterministic, biased component of the process noise. If $G_k = 0$ then we are assuming that the process noise is zero-mean and we get Kalman filter performance. As G_k increases we are assuming that there is more of a deterministic, biased component to the process noise. This gives us better worst-case error performance but worse RMS error performance.
- (b) Compute the next state estimate as follows:

$$\begin{aligned} V_k &= D_k^T D_k \\ \Sigma_k &= (P_k H_k^T H_k - P_k G_k^T G_k + I)^{-1} P_k \\ P_{k+1} &= (I - V_{k+1}) F_k \Sigma_k F_k^T (I - V_{k+1}) + Q_k \\ K_k &= (I - V_{k+1}) F_k \Sigma_k H_k^T \\ \hat{x}_{k+1} &= F_k \hat{x}_k + K_k (y_k - H_k \hat{x}_k) \end{aligned} \quad (12.50)$$

- (c) Verify that

$$(I - G_k P_k G_k^T) \geq 0 \quad (12.51)$$

If not then the filter is invalid.

■ EXAMPLE 12.3

Consider a land-based vehicle that is equipped to measure its latitude and longitude (e.g., through the use of a GPS receiver). This is the same problem as that considered in Example 7.12. The vehicle dynamics and measurements can be approximated by the following equations:

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 0 \\ T \sin \alpha \\ T \cos \alpha \end{bmatrix} u_k + w_k + \delta_k \\ y'_k &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x_k + v'_k \end{aligned} \quad (12.52)$$

The first two components of x_k are the latitude and longitude positions, the last two components of x_k are the latitude and longitude velocities, w_k represents zero-mean process noise due to potholes and other disturbances, δ_k is additional unknown process noise, and u_k is the commanded acceleration. T is the sample period of the system, and α is the heading angle (measured counterclockwise from due east). The measurement y'_k consists of latitude and longitude, and v'_k is the measurement noise. Suppose the standard deviations of the measurement noises are known to be σ_1 and σ_2 . Then we must normalize our measurement equation to satisfy the condition that the measurement noise has a unity covariance. We therefore define the normalized measurement y_k as

$$y_k = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}^{-1} y'_k \quad (12.53)$$

In our simulation we set the covariances of the process and measurement noise as follows:

$$\begin{aligned} Q &= \text{Diag}(4 \text{ m}^2, 4 \text{ m}^2, 1 \text{ (m/s)}^2, 1 \text{ (m/s)}^2) \\ R &= \text{Diag}(\sigma_1^2, \sigma_2^2) = \text{Diag}(900 \text{ m}^2, 900 \text{ m}^2) \end{aligned} \quad (12.54)$$

We can use an H_∞ filter to estimate the position of the vehicle. There may be times when the vehicle is traveling off-road, or on an unknown road, in which case the problem is unconstrained. At other times it may be known that the vehicle is traveling on a given road, in which case the state estimation problem is constrained. For instance, if it is known that the vehicle is traveling on a straight road with a heading of α then the matrix D_k and the vector d_k of Equation (12.32) can be given as follows:

$$\begin{aligned} D_k &= \begin{bmatrix} 1 & -\tan \alpha & 0 & 0 \\ 0 & 0 & 1 & -\tan \alpha \end{bmatrix} \\ d_k &= \begin{bmatrix} 0 & 0 \end{bmatrix}^T \end{aligned} \quad (12.55)$$

We can enforce the condition $D_k D_k^T = I$ by dividing D_k by $\sqrt{1 + \tan^2 \alpha}$. In our simulation we set the sample period T to 1 s and the heading angle α to a constant 60 degrees. The commanded acceleration is toggled between

$\pm 10 \text{ m/s}^2$, as if the vehicle were accelerating and decelerating in traffic. The initial conditions are set to

$$x_0 = [0 \quad 0 \quad 173 \quad 100]^T \quad (12.56)$$

We found via tuning that a G_k matrix of θI , with $\theta = 1/40$, gave good filter performance. Smaller values of θ make the H_∞ filter perform like a Kalman filter. Larger values of θ prevent the H_∞ filter from finding a solution as the positive definite conditions in Equations (12.44) and (12.45) are not satisfied.

This example could be solved by reducing the system-model parameterization [Wen92], or by introducing artificial perfect measurements into the problem [Hay98, Por88]. In fact, those methods could be used for any estimation problem with equality constraints. However, those methods cannot be extended to inequality constraints, whereas the method discussed in this section can be extended to inequality constraints, as discussed earlier.

The unconstrained and constrained H_∞ filters were simulated 100 times each, and the average RMS position and estimation error magnitudes at each time step are plotted in Figure 12.4. It can be seen that the constrained filter results in more accurate estimates. The unconstrained estimator results in position errors that average 35.3 m, whereas the constrained estimator gives position errors that average about 27.1 m. The unconstrained velocity estimation error is 12.9 m/s, whereas the constrained velocity estimation error is 10.9 m/s.

Table 12.1 shows a comparison of the unconstrained and constrained Kalman and H_∞ filters when the noise statistics are nominal. Table 12.2 shows a comparison of the unconstrained and constrained Kalman and H_∞ filters when the acceleration noise on the system has a bias of 1 m/s^2 in both the north and east directions. In both situations, the H_∞ filter estimates position more accurately, but the Kalman filter estimates velocity more accurately. In the off-nominal noise case, the advantage of the H_∞ filter over the Kalman filter for position estimation is more pronounced than when the noise is nominal.

Table 12.1 Example 12.3 estimation errors (averaged over 100 Monte Carlo simulations) of the unconstrained and constrained Kalman and H_∞ filters with nominal noise statistics. The H_∞ filters perform better for position estimation, and the Kalman filters perform better for velocity estimation. Position errors are in units of meters, and velocity errors are in units of meters/second.

	Kalman		H_∞	
	Pos.	Vel.	Pos.	Vel.
Unconstrained	40.3	12.4	35.3	12.9
Constrained	33.2	10.4	27.1	10.9